

Student Information System

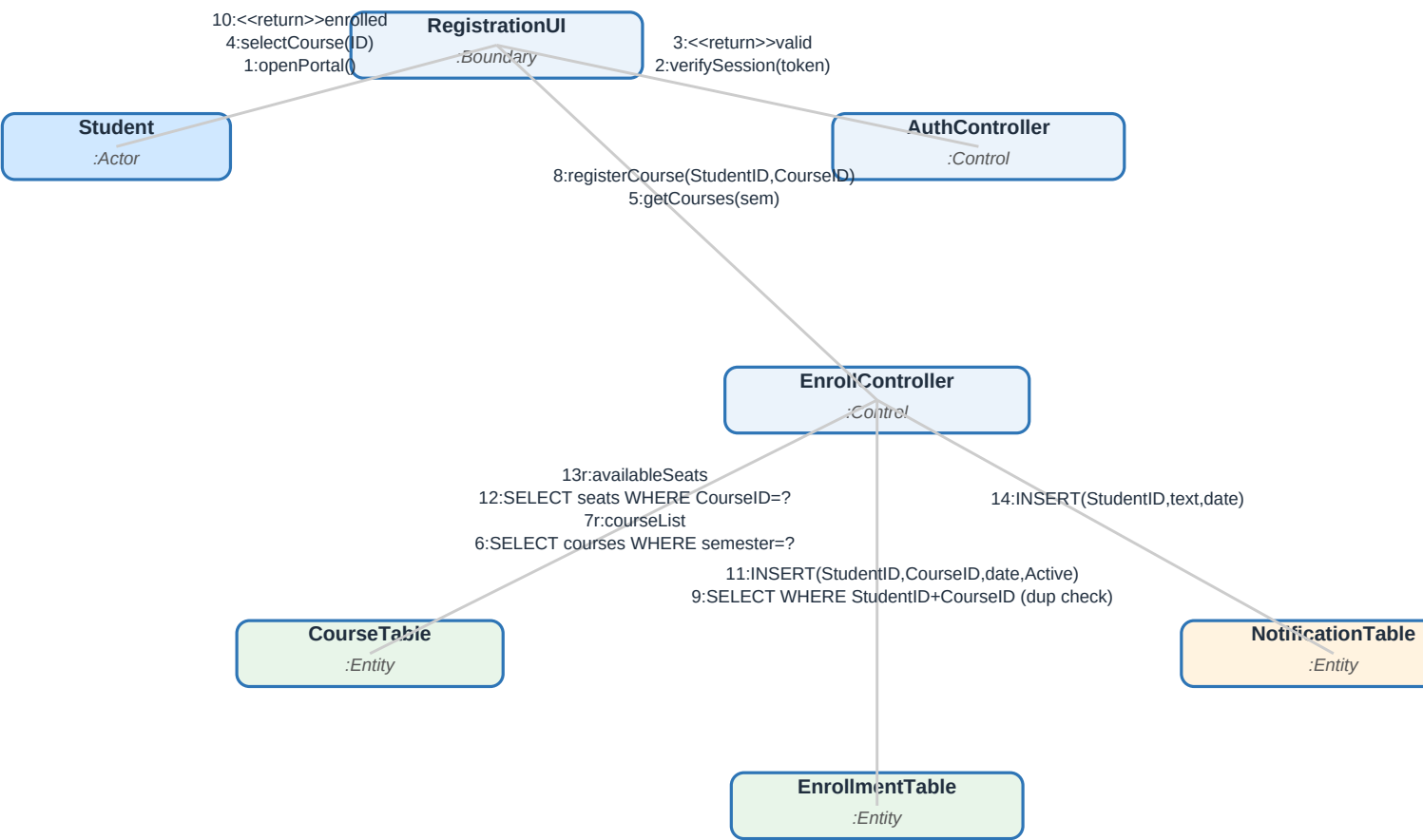
Communication / Collaboration Diagrams

Derived from Sequence Diagrams — All Use Cases

Contents

CC-01	Register for Course	Page 1
CC-02	Login / Authenticate	Page 2
CC-03	Drop Course	Page 3
CC-04	View Grades	Page 4
CC-05	Assign Grade	Page 5
CC-06	Generate Report	Page 6
CC-07	Update Profile	Page 7
CC-08	Add Student	Page 8

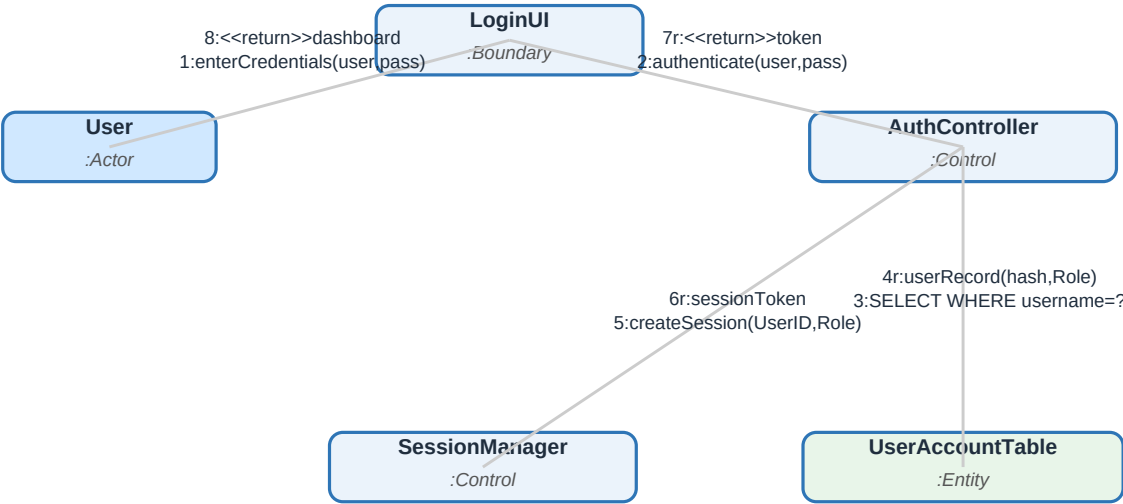
CC-01 — Objects and Collaborations (derived from SC-01)



Message Summary:
Student <-> UI 3 interactions
UI <-> AuthCtrl 1 session check
UI <-> EnrollCtrl 2 calls
EnrollCtrl <-> CourseTable 2 SELECT
EnrollCtrl <-> EnrollmentTbl 1 dup check + 1 INSERT
EnrollCtrl <-> NotifTbl 1 INSERT

ALT: Already enrolled
-> return 'Already registered'
ALT: No seats -> 'Course full'

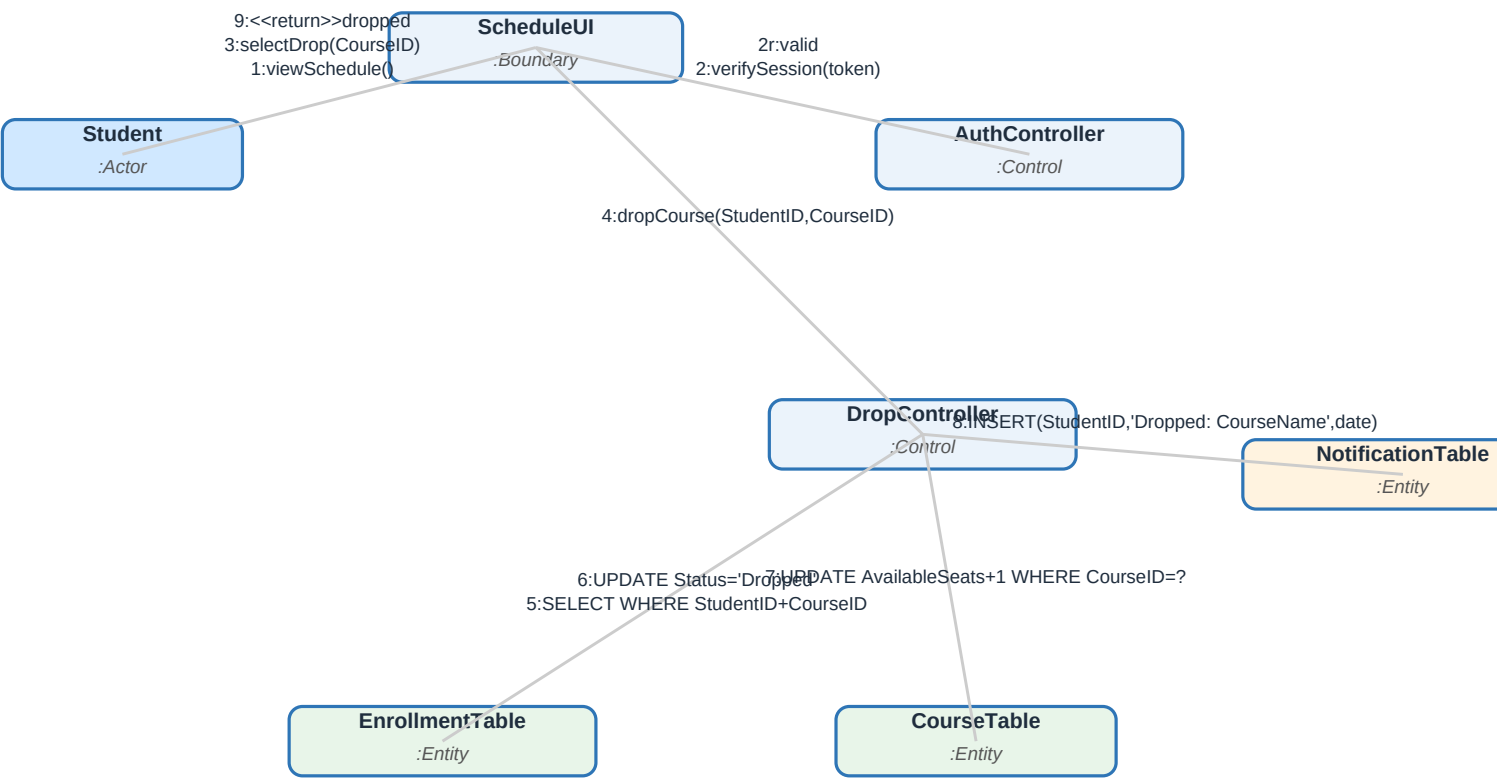
CC-02 — Objects and Collaborations (derived from SC-02)



Message Summary:
User <-> LoginUI 1 interaction
UI <-> AuthCtrl 1 call + return
AuthCtrl <-> UserTbl 1 SELECT
AuthCtrl <-> SessionMgr 1 createSession

ALT: Wrong credentials
-> return 'Invalid username/password'
ALT: Account locked -> show lock msg

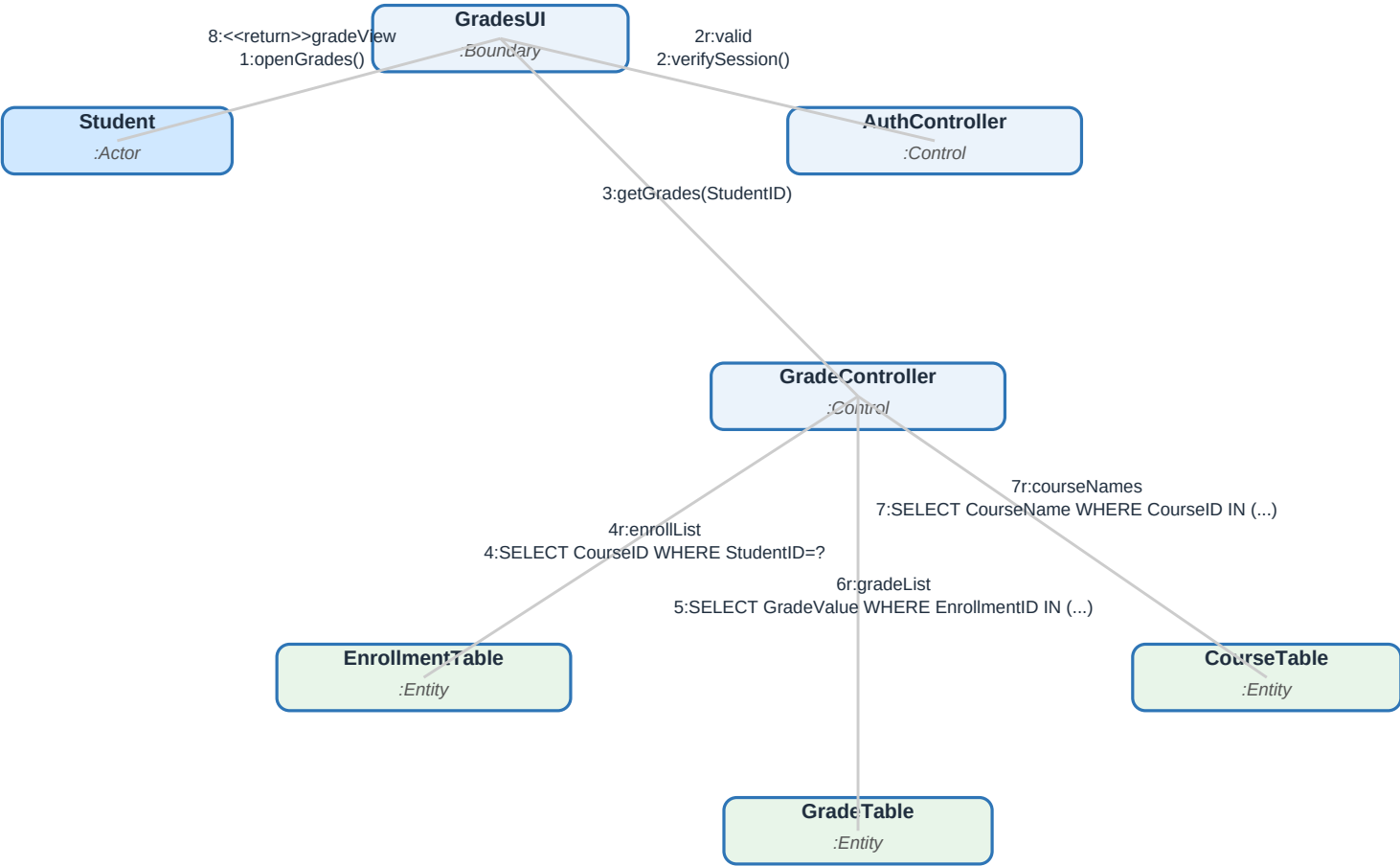
CC-03 — Objects and Collaborations (derived from SC-03)



Message Summary:
Student <-> UI 3 interactions
UI <-> AuthCtrl 1 session check
UI <-> DropCtrl 1 call
DropCtrl <-> EnrollTbl 1 SELECT + 1 UPDATE
DropCtrl <-> CourseTbl 1 UPDATE (seats+1)
DropCtrl <-> NotifTbl 1 INSERT

ALT: Not enrolled -> 'Not enrolled in course'
ALT: Drop deadline passed -> show error

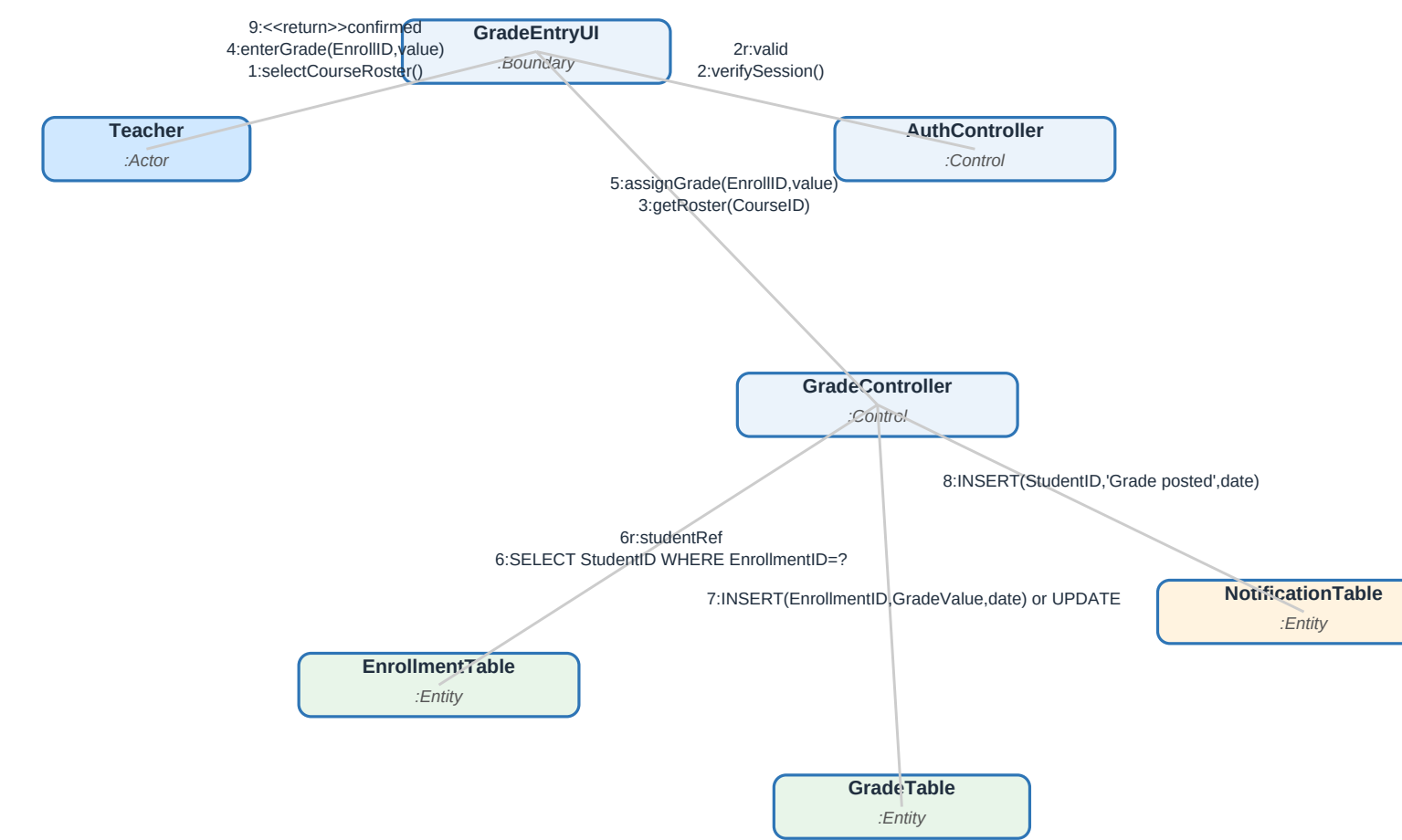
CC-04 — Objects and Collaborations (derived from SC-04)



Message Summary:
Student <-> UI 1 interaction
UI <-> GradeCtrl 1 call
GradeCtrl <-> EnrollTbl 1 SELECT
GradeCtrl <-> GradeTbl 1 SELECT
GradeCtrl <-> CourseTbl 1 SELECT

ALT: No grades yet -> 'No grades recorded'

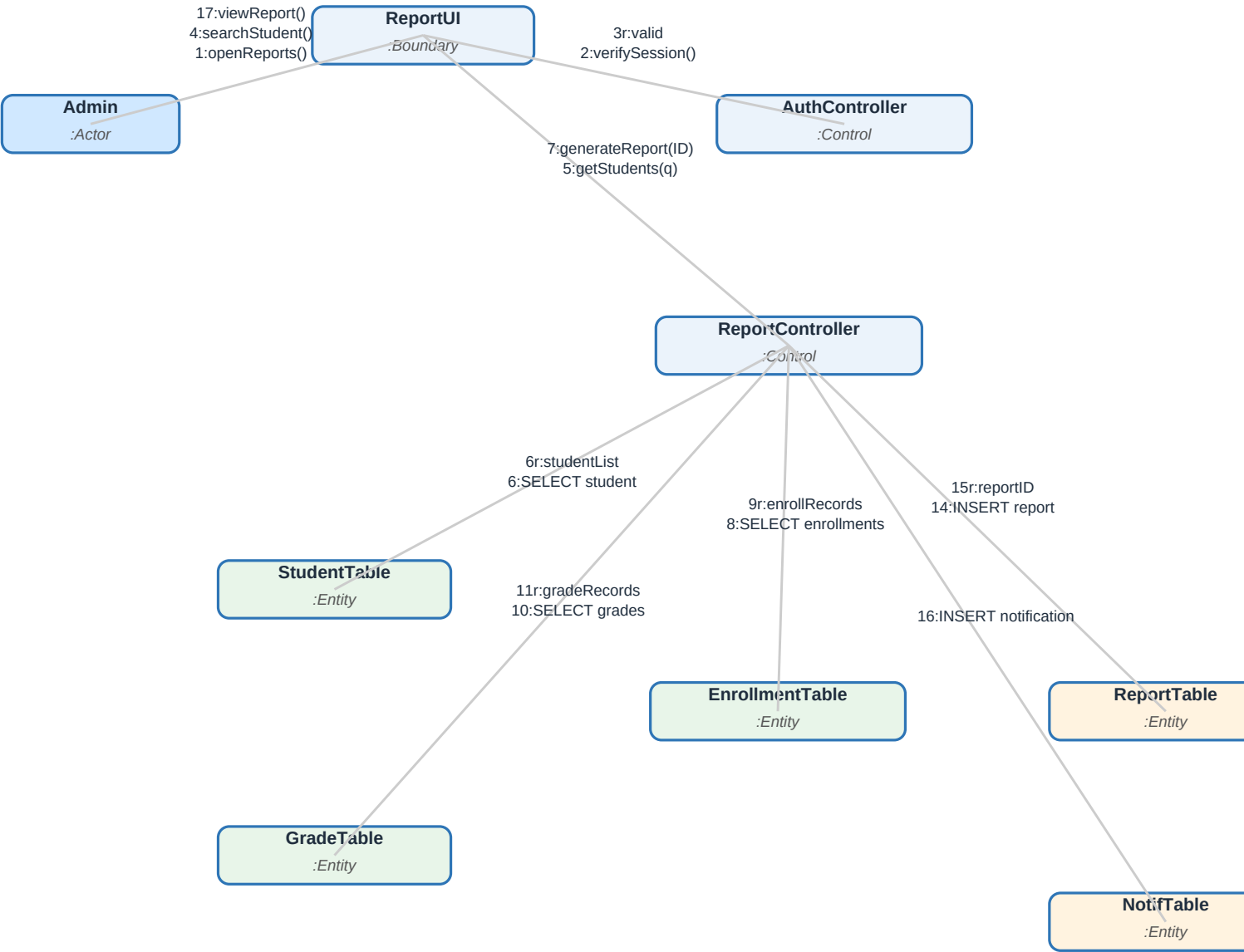
CC-05 — Objects and Collaborations (derived from SC-05)



Message Summary:
Teacher <-> UI 3 interactions
UI <-> AuthCtrl 1 session check
UI <-> GradeCtrl 2 calls
GradeCtrl <-> EnrollTbl 1 SELECT
GradeCtrl <-> GradeTbl 1 INSERT/UPDATE
GradeCtrl <-> NotifTbl 1 INSERT

ALT: Grade already exists -> UPDATE
ALT: Invalid grade value -> show error

CC-06 — Objects and Collaborations (derived from SC-06)

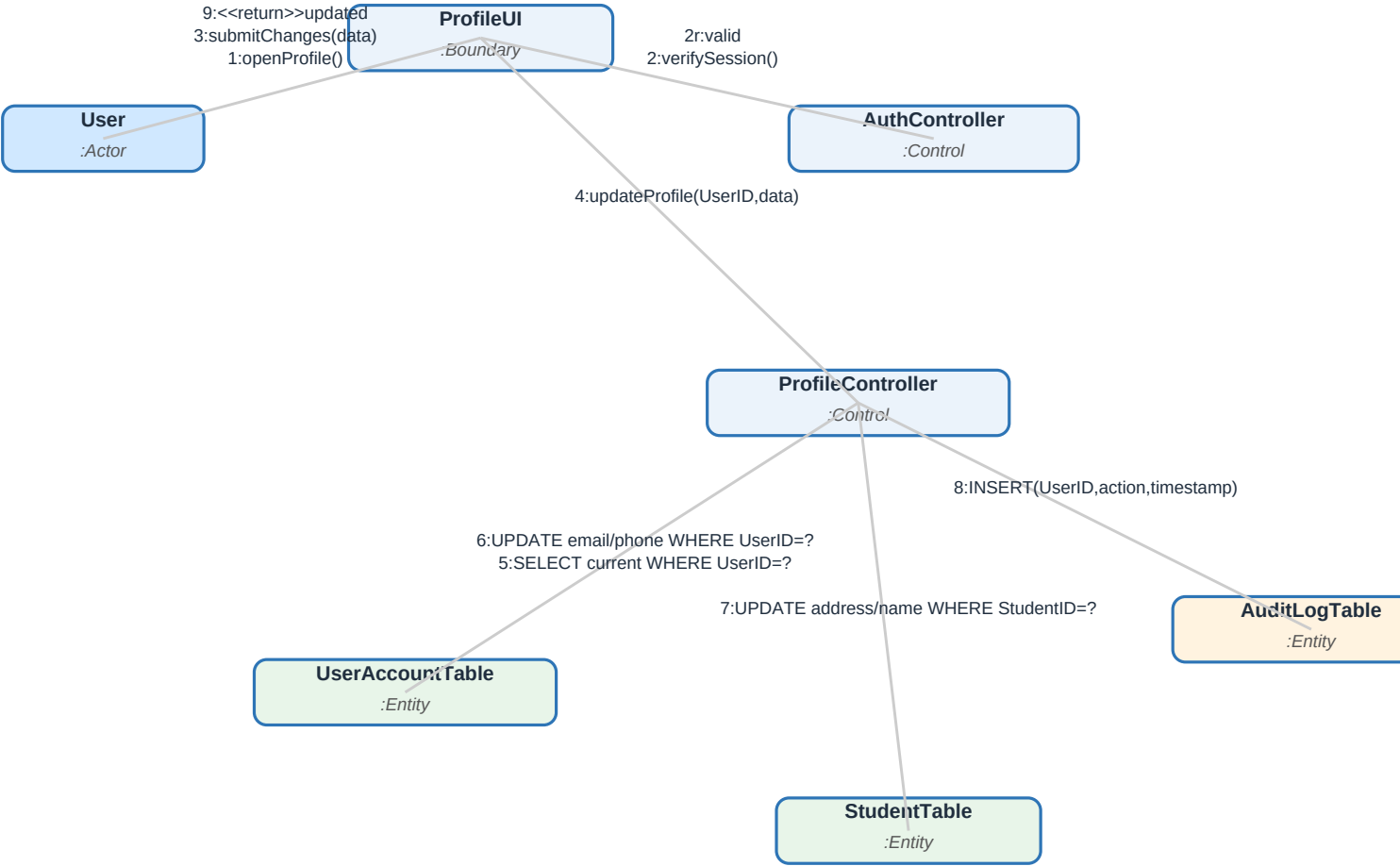


Message Summary:
Admin <-> UI 3 interactions
UI <-> AuthCtrl 1 session check
UI <-> ReportCtrl 2 calls
ReportCtrl <-> StudentTbl 1 SELECT
ReportCtrl <-> EnrollTbl 1 SELECT
ReportCtrl <-> GradeTbl 1 SELECT
ReportCtrl <-> ReportTbl 1 INSERT
ReportCtrl <-> NotifTbl 1 INSERT

loop
12:SELECT CourseName,Credits
13:computeGPA(credits,grades)

ALT: No records -> 'No enrollment data'

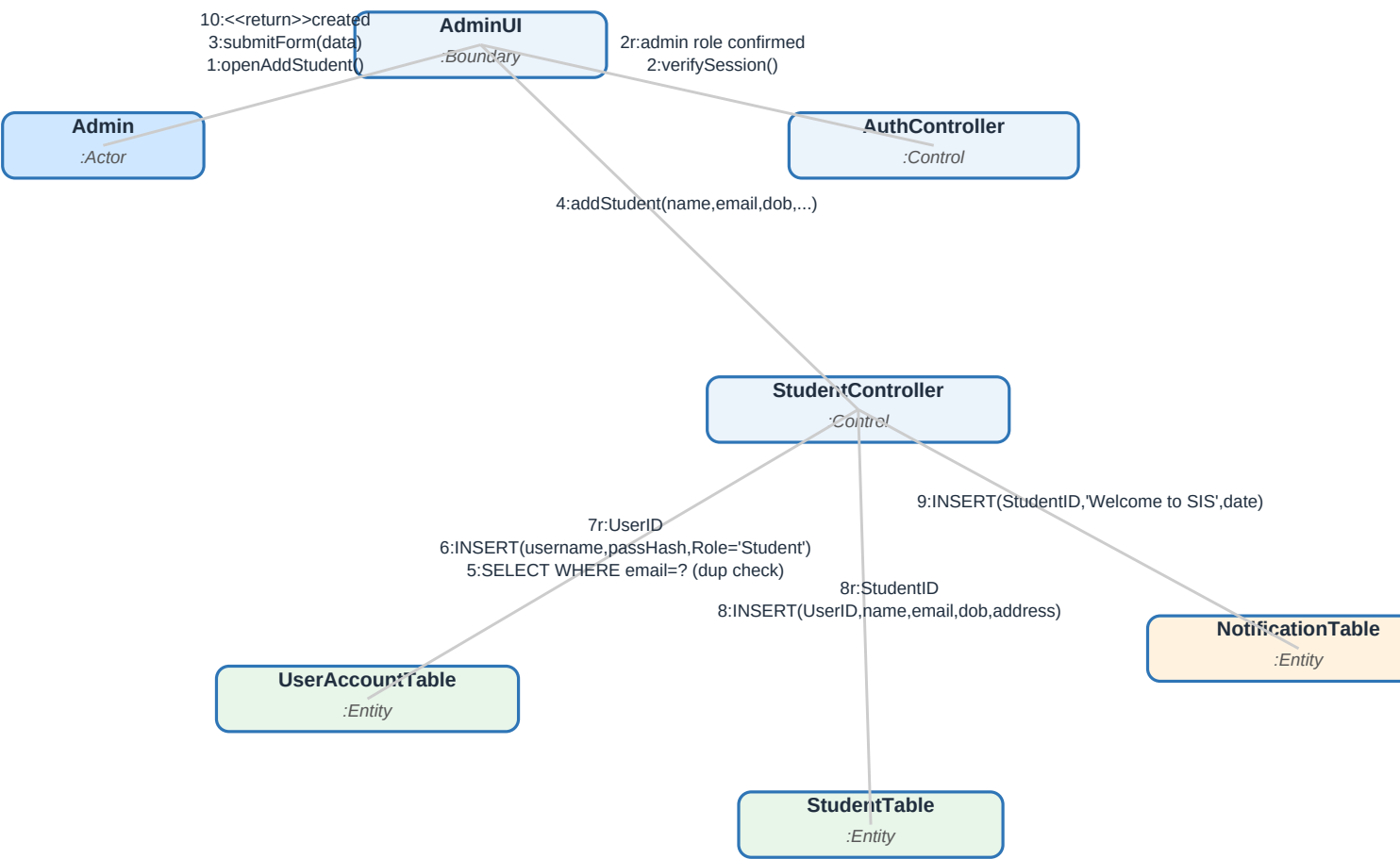
CC-07 — Objects and Collaborations (derived from SC-07)



Message Summary:
User <-> UI 3 interactions
UI <-> AuthCtrl 1 session check
UI <-> ProfileCtrl 1 call
ProfileCtrl <-> UserTbl 1 SELECT + 1 UPDATE
ProfileCtrl <-> StudentTbl 1 UPDATE
ProfileCtrl <-> AuditLog 1 INSERT

ALT: No changes made -> skip DB write
ALT: Validation error -> show field error

CC-08 — Objects and Collaborations (derived from SC-08)



Message Summary:
Admin <-> UI 3 interactions
UI <-> AuthCtrl 1 role check
UI <-> StudentCtrl 1 call
StudentCtrl <-> UserTbl 1 dup check + 1 INSERT
StudentCtrl <-> StudentTbl 1 INSERT
StudentCtrl <-> NotifTbl 1 INSERT (welcome)

ALT: Email already exists
-> return 'Email already registered'
ALT: Missing required field -> validation error